

Cheat sheet

Convert CentOS Linux to RHEL

Convert2RHEL is a command-line utility that facilitates a smooth migration from CentOS Linux 7 to a fully supported Red Hat Enterprise Linux (RHEL) operating system.

Convert2RHEL simplifies the process by converting existing CentOS systems to RHEL, ensuring compatibility and a seamless transition. Because CentOS Linux was derived from RHEL sources, users migrating to RHEL can continue to use many of the same techniques and elements with minimal training, while also gaining more features, tools, support, and value.

This cheat sheet outlines the steps to convert a CentOS Linux instance to RHEL using the Convert2RHEL tool.

1. Validate CentOS

Check whether you have a valid CentOS release version:

```
cat /etc/centos-release
```

Output:

```
CentOS Linux release 7.9.2009 (Core)
```

Switch to root user by using following command

```
sudo su -
```

2. Update your system

Check that your system is up to date before starting the conversion.

```
yum update -y
```

Output:

```
Loaded plugins: fastestmirror
Loading mirror speeds from cached hostfile
* base: centos.excellmedia.net
* extras: centos.excellmedia.net
* updates: centos.excellmedia.net
Resolving Dependencies
-> Running transaction check
-> Package NetworkManager.x86_64 1:1.18.8-1.el7 will be updated
-> Package NetworkManager.x86_64 1:1.18.8-2.el7_9 will be an update
```

3. Secure the packages

Download the signing keys to ensure that we are pulling in only Red Hat created and vetted packages:

```
curl -o /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release  
https://www.redhat.com/security/data/fd431d51.txt
```

Copy the SSL certificates from Red Hat Subscription Management, which will allow us to pull packages over a secure channel:

```
curl -create-dirs -o /etc/rhsm/ca/redhat-uep.pem  
https://ftp.redhat.com/redhat/convert2rhel/redhat-uep.pem
```

4. Download the repository definition

Download a repository definition from the Red Hat FTP server to your `/etc/yum.repos.d/` directory:

```
curl -o /etc/yum.repos.d/convert2rhel.repo  
https://ftp.redhat.com/redhat/convert2rhel/7/convert2rhel.repo
```

To verify Convert2RHEL for CentOS 7 is configured for usage run, enter the following:

```
yum repoinfo convert2rhel-for-rhel-7-rpms
```

Output:

```
Loaded plugins: fastestmirror  
Loading mirror speeds from cached hostfile  
* base: centos.excellmedia.net  
* extras: centos.excellmedia.net  
* updates: centos.excellmedia.net  
convert2rhel-for-rhel-7-rpms | 3.8 kB 00:00  
(1/3): convert2rhel-for-rhel-7-rpms/updateinfo | 4.3 kB 00:00  
(2/3): convert2rhel-for-rhel-7-rpms/primary_db | 4.8 kB 00:00  
(3/3): convert2rhel-for-rhel-7-rpms/group | 124 B 00:02  
Repo-id : convert2rhel-for-rhel-7-rpms  
Repo-name : Convert2RHEL for OS 7  
Repo-status : enabled  
Repo-revision: 1672752339  
Repo-updated : Tue Jan 3 18:55:39 2023  
Repo-pkgs : 7  
Repo-size : 1.0 M  
Repo-baseurl :  
https://cdn.redhat.com/content/public/convert2rhel/7/x86_64/os/  
Repo-expire : 21,600 second(s) (last: Mon Feb 13 18:06:07 2023)  
Filter : read-only:present  
Repo-filename: /etc/yum.repos.d/convert2rhel.repo  
  
repolist: 7
```

5. Install convert2rhel

Install the `convert2rhel` utility:

```
yum install -y convert2rhel
```

Output:

```
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : efibootmgr-17-2.el7.x86_64                1/3
  Installing : pexpect-2.3-11.el7.noarch                 2/3
  Installing : convert2rhel-1.1-1.el7.noarch             3/3
  Verifying  : convert2rhel-1.1-1.el7.noarch             1/3
  Verifying  : pexpect-2.3-11.el7.noarch                2/3
  Verifying  : efibootmgr-17-2.el7.x86_64               3/3

Installed:
  convert2rhel.noarch 0:1.1-1.el7

Dependency Installed:
  efibootmgr.x86_64 0:17-2.el7                pexpect.noarch 0:2.3-11.el7

Complete!
```

6. Convert to RHEL

Find the `organization_id` by running this command:

```
subscription-manager orgs
```

Use the name of the `organization_id` in the next step. Begin the conversion by running the following command:

```
convert2rhel --org <organization_id> --activationkey convert2rhel
```

You can pass a `-y` tag to answer yes to all the questions that follow.

Output:

```
Continue with the system conversion? [y/n]: y

Backing up centos-indexhtml-7-9.el7.centos.noarch.
Successfully downloaded the centos-indexhtml-7-9.el7.centos.noarch package.
Backing up centos-logos-70.0.6-3.el7.centos.noarch.
Successfully downloaded the centos-logos-70.0.6-3.el7.centos.noarch package.
Backing up libreport-centos-2.1.11-53.el7.centos.x86_64.
Successfully downloaded the libreport-centos-2.1.11-53.el7.centos.x86_64
package.
```

```
Backing up libreport-plugin-mantisbt-2.1.11-53.el7.centos.x86_64.  
Successfully downloaded the  
libreport-plugin-mantisbt-2.1.11-53.el7.centos.x86_64 package.  
Removing package: centos-indexhtml-7-9.el7.centos.noarch  
Removing package: centos-logos-70.0.6-3.el7.centos.noarch  
Removing package: libreport-centos-2.1.11-53.el7.centos.x86_64  
Removing package: libreport-plugin-mantisbt-2.1.11-53.el7.centos.x86_64  
...
```

7. Reboot

After the conversion is complete, reboot the system:

```
reboot
```

Take a look at `/etc/redhat-release` confirm it is a RHEL 7 server:

```
cat /etc/redhat-release
```

Output:

```
Red Hat Enterprise Linux Server release 7.9 (Maipo)
```